Are Passive-Dynamic Ankle-Foot Orthoses More Beneficial to Paretic or Non-paretic Limb Energetics for Individuals Post Stroke?

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Introduction

- Individuals post-stroke experience weakened plantar flexor (PF) muscles on paretic side
- PF weakness increases energy cost of walking
- Also necessitates compensatory efforts from non-paretic limb [1]
- Passive-dynamic ankle-foot orthoses (PD-AFOs) often prescribed to support paretic PFs
- Stiffness-personalized PD-AFOs can reduce net mechanical cost of transport (COT) compared to No AFO or standard-of-care (SOC) AFO [2]
- **Definitive cause of net COT reduction for individuals post-stroke with PD-AFO use remains unclear**

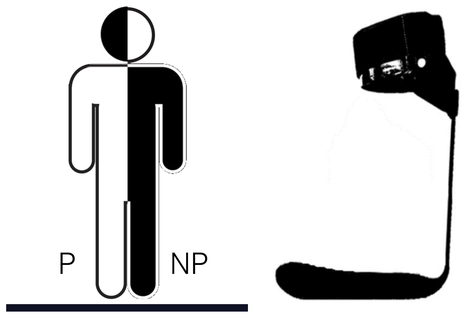

Figure 1: Diagram of paresis caused by stroke (left), and a PD-AFO (right)

Hypothesis

For individuals post-stroke, **reductions in net COT** while wearing a PD-AFO are **primarily driven by reductions in the COT of the paretic limb**

Methods

- Instrumented treadmill and full-body motion capture
- Walked with No AFO, SOC AFO, and personalized PD-AFO at baseline No AFO self-selected walking speed
- COT calculated for both limbs in each condition [3]:

Individuals Post-Stroke (n=20)	
Age	63.5 (10.1) years
Height	1.74 (1.1) m
Mass	88.2 (17.7) kg
Gender	14 M / 6 F

$$COT = \frac{\dfrac{\sum +Work_{Hip,\,Knee,\,Ankle,\,Distal\ Foot} + \left|\sum -Work_{Hip,\,Knee,\,Ankle,\,Distal\ Foot}\right|}{Body\ Mass}}{Stride\ Length}$$

- Skillings Mack test used to compare average COT values across conditions for both limbs (Wilcoxon signed-rank post-hoc tests)

Results

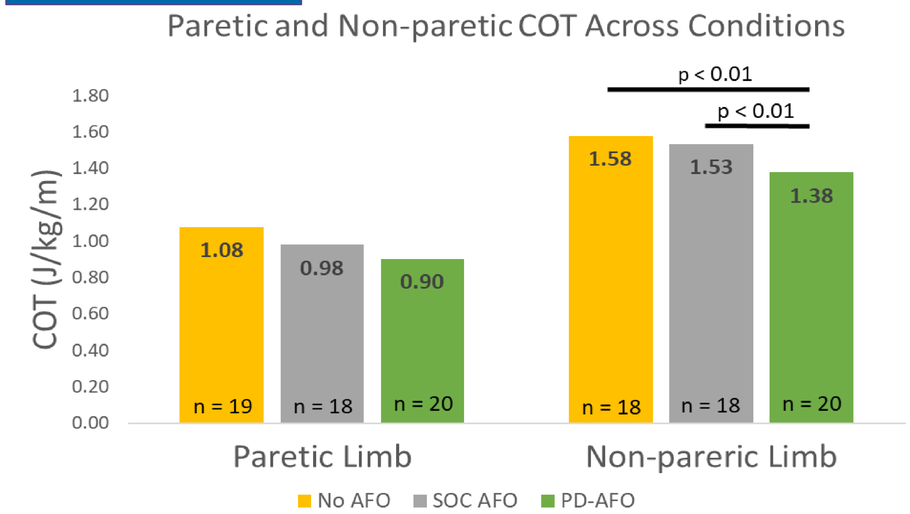

Figure 2: Group means of mechanical cost of transport for paretic and non-paretic limb in each brace condition

Hypothesis Rejected

Secondary Analysis

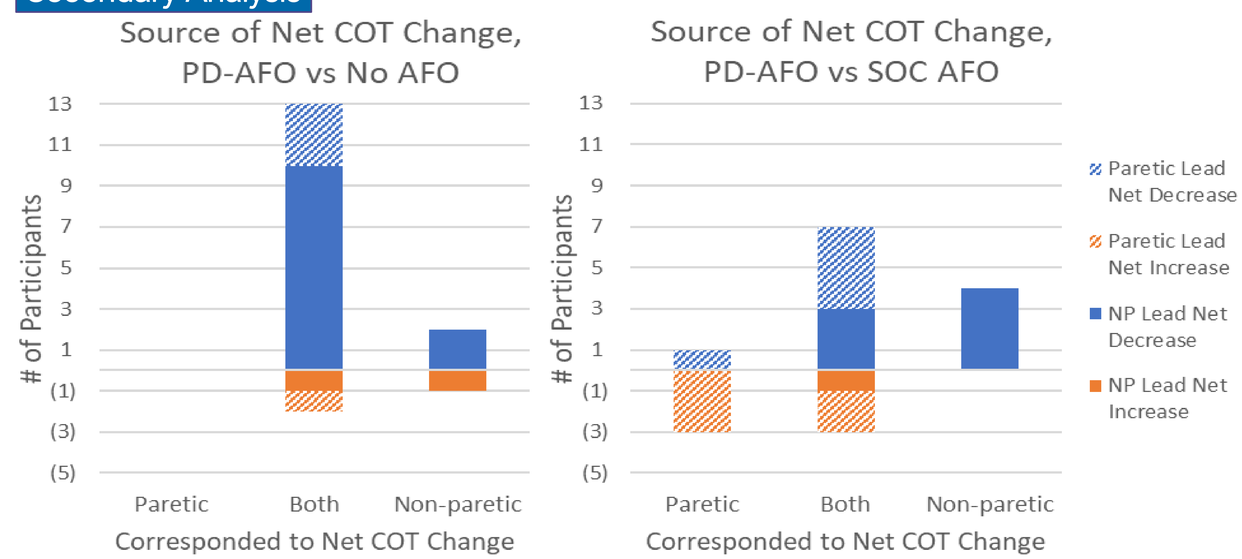

Figure 3: Count of participants whose paretic and non-paretic limb COT changes matched their net COT change while walking with the PD-AFO compared to No AFO (left, n=18) or SOC AFO (right, n=18). Solid blocks represent participants with larger change in non-paretic COT

Significance

While stiffness-customized PD-AFOs can enhance the paretic limb of individuals post-stroke, **a larger benefit for most participants is lessening the compensatory burden on their non-paretic limb**

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When designing and customizing AFOs, and likely other rehabilitation interventions for individuals post-stroke, **it is crucial to consider the effect on the entire lower limb system, especially the the non-paretic limb**

References: [1] Roelkar, S. et al. Gait & Posture, 68, 6-14, 2019; [2] Skigen, J. et al. Preprint, in review; [3] Ebrahimi, A. et al. Gait & Posture, 56, 49-53, 2017

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